

Completing Short Algebraic and Segment Proofs

Answer Key

High School Geometry

Quick Answers

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|----------|----------|----------|--------------|
| 1. 6 | 3. 9 | 5. 6, 8 | 7. 8 |
| 2. 6, 17 | 4. 4, 15 | 6. 8, 28 | 8. 8, 13, 19 |
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Curriculum

Level: High School Geometry

HSA-REI.A.1 – problems 1, 3, 7

HSG-CO.A – problems 4, 6

HSG-CO.C – problems 2, 5, 8

Difficulty 1–5 of 5 across 21 tagged checks.

1. Given $3x + 7 = 25$; prove $x = 6$.

Each line changes the equation, so each line names the property that permits the change.

Statements	Reasons
1. $3x + 7 = 25$	1. Given
2. $3x = 18$	2. Subtraction Property of Equality
3. $x = 6$	3. Division Property of Equality

$$\boxed{x = 6}$$

Manual review: the Reasons column is written work — accept “subtracted 7 from both sides” only when the property is named.

2. B between A and C : $AB = 2x + 5$, $BC = x + 3$, $AC = 26$ cm.

(a) The **Segment Addition Postulate**. Betweenness is what licenses adding the two parts; it is assumed, not proved.

(b) Substitute into $AB + BC = AC$ and collect:

$$(2x + 5) + (x + 3) = 26 \implies 3x + 8 = 26 \implies 3x = 18$$

$$\boxed{x = 6}$$

(c) $AB = 2(6) + 5 = 12 + 5$.

$$\boxed{AB = 17 \text{ cm}}$$

Check: $BC = 6 + 3 = 9$ cm, and $17 + 9 = 26$ cm ✓

Manual review: part (a) is a named justification.

3. Given $5(x - 2) = 3x + 8$; **prove** $x = 9$.

Statements	Reasons
1. $5(x - 2) = 3x + 8$	1. Given
2. $5x - 10 = 3x + 8$	2. Distributive Property
3. $2x - 10 = 8$	3. Subtraction Property of Equality
4. $2x = 18$	4. Addition Property of Equality
5. $x = 9$	5. Division Property of Equality

$$x = 9$$

Grading note: line 3 must remove $3x$ from *both* sides. A student who writes $2x - 10 = 8$ with the reason “combined like terms” has the right statement and the wrong reason — the terms are on opposite sides of the equals sign, so nothing is being combined; a quantity is being subtracted from both sides. *Manual review:* read the *Reasons* column.

4. M is the midpoint of $\overline{AB}$: $AM = 4x - 1$, $MB = 2x + 7$ **cm.**

(a) The proof runs from the definition of midpoint to a numerical equation, then solves it:

Statements	Reasons
1. M is the midpoint of $\overline{AB}$	1. Given
2. $\overline{AM} \cong \overline{MB}$	2. Definition of midpoint
3. $AM = MB$	3. Definition of congruent segments
4. $4x - 1 = 2x + 7$	4. Substitution
5. $2x - 1 = 7$	5. Subtraction Property of Equality
6. $2x = 8$	6. Addition Property of Equality
7. $x = 4$	7. Division Property of Equality

(b) $x = 4$, so $AM = 4(4) - 1 = 15$.

$$AM = 15 \text{ cm}$$

Check: $MB = 2(4) + 7 = 15$ cm, equal as the midpoint requires ✓

Manual review: lines 2 and 3 are the two separate facts students most often merge — midpoint gives congruence, and congruence gives equal *measures*.

5. A, B, C, D collinear in order: $AB = 3x$, $BC = x + 2$, $CD = 2x - 1$, $AD = 37$ **cm.**

(a) The **Segment Addition Postulate**, applied twice, gives $AB + BC + CD = AD$:

$$3x + (x + 2) + (2x - 1) = 37 \implies 6x + 1 = 37 \implies 6x = 36$$

$$x = 6$$

(b) $BC = 6 + 2$.

$$BC = 8 \text{ cm}$$

Check: $AB = 18$, $BC = 8$, $CD = 11$ cm, and $18 + 8 + 11 = 37$ cm ✓

6. $\overline{AB} \cong \overline{CD}$: $AB = 3x + 4$, $CD = 5x - 12$ cm.

(a)

Statements	Reasons
1. $\overline{AB} \cong \overline{CD}$	1. Given
2. $3x + 4 = 5x - 12$	2. Definition of congruent segments
3. $4 = 2x - 12$	3. Subtraction Property of Equality
4. $16 = 2x$	4. Addition Property of Equality
5. $x = 8$	5. Division Property of Equality

$$x = 8$$

(b) $AB = 3(8) + 4 = 24 + 4$.

$$AB = 28 \text{ cm}$$

Check: $CD = 5(8) - 12 = 28$ cm, so the two segments really are congruent ✓ *Manual review: read the Reasons column.*

7. Find Jordan's mistake in $2(x + 3) = 4x - 10$.

Line 3 is the first wrong line. Jordan subtracted $2x$ from the left side only. The Subtraction Property of Equality permits subtracting the same quantity from *both* sides; taking it from one side changes the equation into a different one, so every line after it is about the wrong equation.

The corrected proof:

Statements	Reasons
1. $2(x + 3) = 4x - 10$	1. Given
2. $2x + 6 = 4x - 10$	2. Distributive Property
3. $6 = 2x - 10$	3. Subtraction Property of Equality
4. $16 = 2x$	4. Addition Property of Equality
5. $x = 8$	5. Division Property of Equality

$$x = 8$$

Check: $2(8 + 3) = 22$ and $4(8) - 10 = 22$ ✓. Jordan's $x = 4$ fails this check: $2(7) = 14$ but $4(4) - 10 = 6$. *Manual review: the student must name the line and the property.*

8. Synthesis: $\overline{AB} \cong \overline{CD}$ forces $\overline{AC} \cong \overline{BD}$.

(a) The two long segments share $\overline{BC}$, so add it to each of the congruent short segments:

Statements	Reasons
1. A, B, C, D collinear in order; $\overline{AB} \cong \overline{CD}$	1. Given
2. $AB = CD$	2. Definition of congruent segments
3. $AB + BC = CD + BC$	3. Addition Property of Equality
4. $AB + BC = AC$; $BC + CD = BD$	4. Segment Addition Postulate
5. $AC = BD$	5. Substitution
6. $\overline{AC} \cong \overline{BD}$	6. Definition of congruent segments

(b) $2x - 3 = x + 5$, so subtracting x from both sides and adding 3 to both sides gives

$$\boxed{x = 8}$$

Then $AB = 2(8) - 3 = 16 - 3$, which is $\boxed{AB = 13 \text{ cm}}$, and $CD = 8 + 5 = 13 \text{ cm}$ as the congruence requires.

Finally $AC = AB + BC = 13 + 6$.

$$\boxed{AC = 19 \text{ cm}}$$

Check: $BD = BC + CD = 6 + 13 = 19 \text{ cm}$, equal to AC exactly as part (a) proved ✓

Manual review: read the student's two-column proof in part (a).